$$\mathcal{A}_{\Theta,G} : D \mapsto Y$$

$$D \xrightarrow{\Theta_0} X^{(0)} \xrightarrow{\nu} (\widehat{X}^{(0)}, m^{(0)}) \xrightarrow{G} M \xrightarrow{\mathcal{T}_\Theta} Y \xrightarrow{\nu} (\widehat{Y}, m^Y)$$

$$n, d, L, H, d_h, d_v, k \in \mathbb{N}, \qquad \varepsilon > 0$$

$$X^{(0)} = F_{\Theta_0}(D), \qquad X^{(\ell)} \in \mathbb{R}^{n\times d}$$

$$x_i^{(\ell)} = \mathrm{row}_i(X^{(\ell)})^\mathsf{T} \in \mathbb{R}^d, \qquad 0 \le \ell \le L$$

$$m_i^{(\ell)} = \left\| x_i^{(\ell)} \right\|_2 = \sqrt{(x_i^{(\ell)})^\mathsf{T} x_i^{(\ell)}}$$

$$\widehat{x}_i^{(\ell)} = \frac{x_i^{(\ell)}}{m_i^{(\ell)} + \varepsilon}, \qquad \widehat{x}_i^{(\ell)} \in S^{d-1}$$

$$S^{d-1} = \{u \in \mathbb{R}^d : \|u\|_2 = 1\}$$

$$\widehat{X}^{(\ell)} = \begin{bmatrix} (\widehat{x}_1^{(\ell)})^\mathsf{T} \\ (\widehat{x}_2^{(\ell)})^\mathsf{T} \\ \vdots \\ (\widehat{x}_n^{(\ell)})^\mathsf{T} \end{bmatrix} \in \mathbb{R}^{n\times d}$$

$$m^{(\ell)} = \begin{bmatrix} m_1^{(\ell)} \\ m_2^{(\ell)} \\ \vdots \\ m_n^{(\ell)} \end{bmatrix} \in \mathbb{R}^{n\times 1}$$

$$r_i^{(\ell)} = \log(m_i^{(\ell)} + \varepsilon), \qquad r^{(\ell)} = \begin{bmatrix} r_1^{(\ell)} \\ r_2^{(\ell)} \\ \vdots \\ r_n^{(\ell)} \end{bmatrix}$$

$$Z^{(\ell)} = \begin{bmatrix} \widehat{X}^{(\ell)} & r^{(\ell)} \end{bmatrix} \in \mathbb{R}^{n\times(d+1)}$$

$$G = (V_G, E_G), \qquad V_G = \{1, 2, \dots, n\}, \qquad E_G \subseteq V_G \times V_G$$

$$M_{ij} = \mathbf{1}_{(i,j)\in E_G}, \qquad M \in \{0,1\}^{n\times n}$$

$$M_{ij}^+ = \max(M_{ij}, \delta_{ij})$$

$$C_{ij}^{(\ell)} = (\widehat{x}_i^{(\ell)})^\mathsf{T} \widehat{x}_j^{(\ell)}$$

$$\Delta_{ij}^{(\ell)} = \left\| \widehat{x}_i^{(\ell)} - \widehat{x}_j^{(\ell)} \right\|_2^2 = 2(1 - C_{ij}^{(\ell)})$$

$$\mathcal{K}_i^{(\ell)} = \mathrm{TopK}_j(C_{ij}^{(\ell)}, k)$$

$$T_{ij}^{(\ell)} = \mathbf{1}_{j \in \mathcal{K}_i^{(\ell)}}$$

$$\overline{M}_{ij}^{(\ell)} = M_{ij}^{+} T_{ij}^{(\ell)}$$

$$B_{ij}^{(\ell)} = \begin{cases} 0, & \overline{M}_{ij}^{(\ell)} = 1 \\ -\infty, & \overline{M}_{ij}^{(\ell)} = 0 \end{cases}$$

$$W_Q^{(\ell,h)}, W_K^{(\ell,h)} \in \mathbb{R}^{d \times d_h}, \qquad W_V^{(\ell,h)} \in \mathbb{R}^{(d+1) \times d_v}$$

$$W_O^{(\ell)} \in \mathbb{R}^{H d_v \times d}, \qquad W_R^{(\ell)} \in \mathbb{R}^{d \times d}$$

$$Q^{(\ell,h)} = \widehat{X}^{(\ell)} W_Q^{(\ell,h)}$$

$$K^{(\ell,h)} = \widehat{X}^{(\ell)} W_K^{(\ell,h)}$$

$$V^{(\ell,h)} = Z^{(\ell)} W_V^{(\ell,h)}$$

$$S^{(\ell,h)} = \frac{Q^{(\ell,h)} (K^{(\ell,h)})^{\mathsf{T}}}{\sqrt{d_h}} + B^{(\ell)}$$

$$P_{ij}^{(\ell,h)} = \frac{\exp(S_{ij}^{(\ell,h)})}{\sum_{q=1}^{n} \exp(S_{iq}^{(\ell,h)})}$$

$$P^{(\ell,h)} = \mathrm{softmax}_j(S^{(\ell,h)})$$

$$P_{ij}^{(\ell,h)} = 0 \qquad \forall\, \overline{M}_{ij}^{(\ell)} = 0$$

$$O^{(\ell,h)} = P^{(\ell,h)} V^{(\ell,h)}$$

$$O^{(\ell)} = \begin{bmatrix} O^{(\ell,1)} & O^{(\ell,2)} & \cdots & O^{(\ell,H)} \end{bmatrix} W_O^{(\ell)}$$

$$R^{(\ell)} = X^{(\ell)} W_R^{(\ell)} + O^{(\ell)}$$

$$X^{(\ell+1)} = R^{(\ell)}$$

$$Y = X^{(L)}$$

$$y_i = \mathrm{row}_i(Y)^{\mathsf{T}}, \qquad m_i^Y = \|y_i\|_2\,, \qquad \widehat{y}_i = \frac{y_i}{m_i^Y + \varepsilon}$$

$$\widehat{Y} = \begin{bmatrix} \widehat{y}_1^\mathsf{T} \\ \widehat{y}_2^\mathsf{T} \\ \vdots \\ \widehat{y}_n^\mathsf{T} \end{bmatrix}, \qquad m^Y = \begin{bmatrix} m_1^Y \\ m_2^Y \\ \vdots \\ m_n^Y \end{bmatrix}$$

$$R^\star \in \mathbb{R}^{n \times d}, \qquad r_i^\star = \mathrm{row}_i(R^\star)^\mathsf{T}$$

$$m_i^\star = \|r_i^\star\|_2, \qquad \widehat{r}_i^\star = \frac{r_i^\star}{m_i^\star + \varepsilon}$$

$$\mathcal{L}_1 = \sum_{i=1}^{n} \left(1 - \widehat{y}_i^\mathsf{T} \widehat{r}_i^\star\right)$$

$$\mathcal{L}_2 = \sum_{i=1}^{n} \left(\log(m_i^Y + \varepsilon) - \log(m_i^\star + \varepsilon)\right)^2$$

$$\mathcal{L}_3 = \sum_{\ell=0}^{L-1} \sum_{h=1}^{H} \left\| P^{(\ell,h)} \odot (\mathbf{1}\mathbf{1}^\mathsf{T} - \overline{M}^{(\ell)}) \right\|_F^2$$

$$\mathcal{L}_4 = \sum_{\ell=0}^{L-1} \left\| X^{(\ell+1)} - X^{(\ell)} \right\|_F^2$$

$$\mathcal{L}_\Theta = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2 + \lambda_3 \mathcal{L}_3 + \lambda_4 \mathcal{L}_4$$

$$\Theta_{t+1} = \Theta_t - \eta \nabla_\Theta \mathcal{L}_\Theta$$

$$X_{t+1}^{(0)} = Y_t$$

$$D \to X \to (\widehat{X}, m) \to G \to M \to (Q, K, V) \to P \to Y \to (\widehat{Y}, m^Y)$$

$$\boxed{\mathcal{A}_{\Theta,G} = \nu \circ \mathcal{T}_\Theta \circ M_G \circ \nu \circ F_{\Theta_0}}$$